Efficiency and resource measurement

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make faster a measured statement under a declared protocol.

Teach eligibility, percentages and storage with complete checked examples.

Keep one habit: Model ratios → remove help → assess new timings and layer widths → plan fair benchmarking.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Calculate **throughput** and amortised time per example from repeated timings

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Different examples or group proportions can change accuracy and workload. Shared cases and declared conditions help isolate model differences rather than differences in the evaluation sample.

Opening retrieval

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain how a changed **input** mix can change overall accuracy

Reference example, not the only acceptable wording: Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both.

Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim. The mix changes while observed group rates remain the same.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a reused feature extractor and a newly fitted **output** head

Reference example, not the only acceptable wording: Teacher explanation: The extractor gives h = 6 and the head predicts 1.

Weights [1, 1] and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or calibration.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Look for: Encode an ordered sequence with an explicit vocabulary

Reference example, not the only acceptable wording: Teacher explanation: IDs are [2, 3, 4, 0], mask [1, 1, 1, 0], real length 3 and final real event STOP at position 2.

Pictured end goal

By the end, I can explain efficiency and resource measurement using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Dataset shift in neural models

Transfer learning

Sequence representations

For the compact alternative schedule, use two 45-minute sessions. The timings are invented for arithmetic and do not describe a particular device or trained model. Keep units and denominators visible.

Separate a validation choice from a general performance claim, and separate raw parameter payload from file size, runtime memory and energy.

Outcomes

Calculate **throughput** and amortised time per example from repeated timings

Count dense-network parameters and distinguish storage from **peak memory**

Compare resource use under fixed quality and measurement conditions

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the workload and decision contract before the ratios

> Keep main median timings separate from pictured means and independent cases.

Definition, fixed protocol and complete original timings

A neural model can be faster yet fail the task's accuracy requirement. **Training** cost and **prediction** cost are different: **training** updates parameters, while inference uses the fitted model.

Compare quality and resources together under a declared test, rather than calling a model efficient from one small number.

Use these invented classroom timings for models A and B. Each timed pass **process**es the same 100 pre**process**ed examples sequentially, one at a time, on the same device and precision.

Start before the first **prediction** and stop after the final **prediction** completes. Loading and pre**process**ing are excluded for both. One untimed warm-up pass precedes three timed passes; the warm-up work must still be reported.

The validation cases are the same for both models and are scored once each; repeating timing passes does not create 300 independent quality examples. No actual benchmark was run to produce this table.

Measurement	Model A	Model B
Pass 1 seconds	2.0	1.2
Pass 2 seconds	2.2	1.1
Pass 3 seconds	1.8	1.3
Correct on 100 validation cases	92	90

Checked timing calculations and summary

Sort each set of three times and take the middle value. A's median is 2.0 seconds per 100 examples; B's is 1.2 seconds. Dividing by 100 gives amortised times of 0.020 and 0.012 seconds per example, or 20 and 12 milliseconds.

Median-pass **throughput** is 100 / 2.0 = 50 examples per second for A and about 83.33 for B.

The code assumes a positive integer example count and numeric times. A **output**s (2.0, 20.0, 50.0); B **output**s approximately (1.2, 12.0, 83.33). Retain full precision until display.

The time reduction is (20 - 12) / 20 = 40 percent, while **throughput** rises by about 66.67 percent. These are different denominators.

Amortised time does not reveal the slowest request or a **latency** percentile. In a batched or concurrent workload, total time divided by count is not an individual request’s waiting time.

Record per-request timings if the requirement concerns **latency**; keep batch size and concurrency explicit.

Derived quantity	A	B
Sorted pass seconds	1.8, 2.0, 2.2	1.1, 1.2, 1.3
Median seconds per 100 cases	2	1.2
Amortised milliseconds per case	20	12
Throughput at median pass, cases/second	50	83.33 approximately
Correct of same 100 validation cases	92	90
At least 91 correct AND at most 25 ms	Pass both	Fail quality
Timed prediction executions	300	300
Warm-up executions, excluded from timing	100	100

Main eligibility and alternative rule key

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

Guided: under the alternative rule set in advance, A and B both qualify. B wins the declared lower-amortised-time tie-breaker, 12 versus 20 ms.

Exit: best depends on requirements and B fails the original quality rule; greener is unsupported without energy **evidence**.

Use identical workload and declared conditions or label a whole-system comparison, measure completed work, repeat timings, retain raw results and quality outcomes, and reserve final evaluation after selection.

The requirements were set before inspecting these results. B is faster under this measurement but is not eligible under the quality rule.

One hundred validation cases do not establish future accuracy; reserve independent final evaluation after choosing a model, and inspect relevant subgroup results.

If the quality requirement had instead been set in advance to at least 90 correct, with the same time limit, which models would qualify? Which would you choose if lower measured amortised time is the declared tie-breaker?

Dense parameter and measurement-scope account

Dense 4 → 3 → 2 network	Weights	Biases	Total parameters
4 → 3 layer	4 × 3 = 12	3	15
3 → 2 layer	3 × 2 = 6	2	8
Whole network	18	5	23
Four bytes per parameter	23 × 4 = 92 raw bytes	Metadata not counted	Activations and buffers not counted
Peak training memory	Requires actual scoped measurement	Gradients / optimizer state may add storage	Not supplied
Energy or carbon emissions	Require additional evidence	Time alone is insufficient	Not supplied

Separate picture and complete inputs

Compare speed and quality under the same case count and acceptance contract.

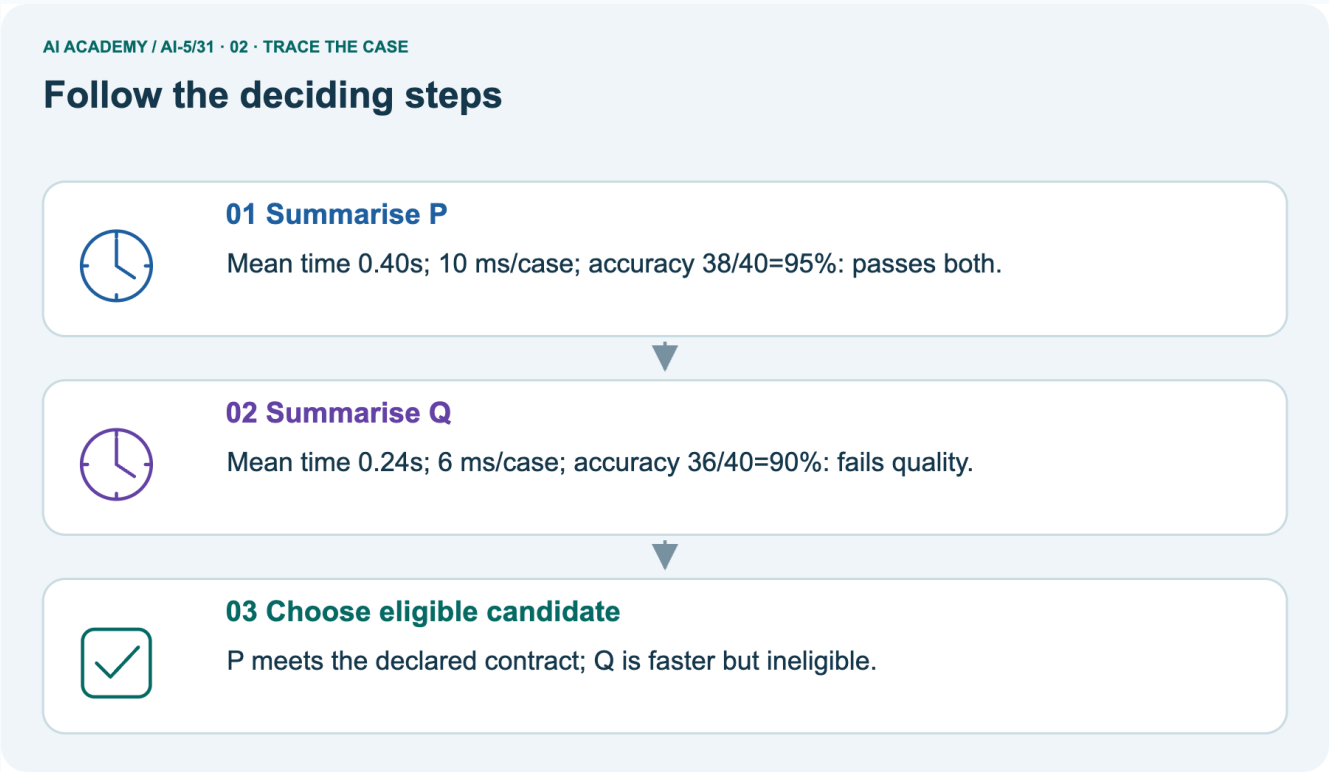
Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Requireaccuracy>=93% and mean amortised time<=12 ms/case.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main comparison uses the median of three 100-case passes. The picture is a separate 40-case comparison using the mean: P times .32, .40, .48 seconds average .40; Q times .20, .24, .28 average .24. That gives 10 and 6 ms per case. Accuracy P 38/40 = 95%, Q 36/40 = 90%. With at least 93% and at most 12 ms required, only P qualifies.

Although these symmetric lists also have matching medians, mean and median are different summaries in general.

Picture trace and changed-contract key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Summarise P	Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both.
2. Summarise Q	Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality.
3. Choose eligible candidate	P meets the declared contract; Q is faster but ineligible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

Change: Lower the accuracy requirement to90% before a new selection exercise.

Prediction: Both become eligible; Q has the smaller measured mean amortised time.

Reason: The acceptance contract changed; Q's measured accuracy did not improve.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Latency	Elapsed time for one request from a declared start to completed response
Throughput	The number of completed examples processed per unit time
Parameter storage	The bytes used to store model weights and biases at a declared precision
Peak memory	The largest amount of memory in use during a measured operation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

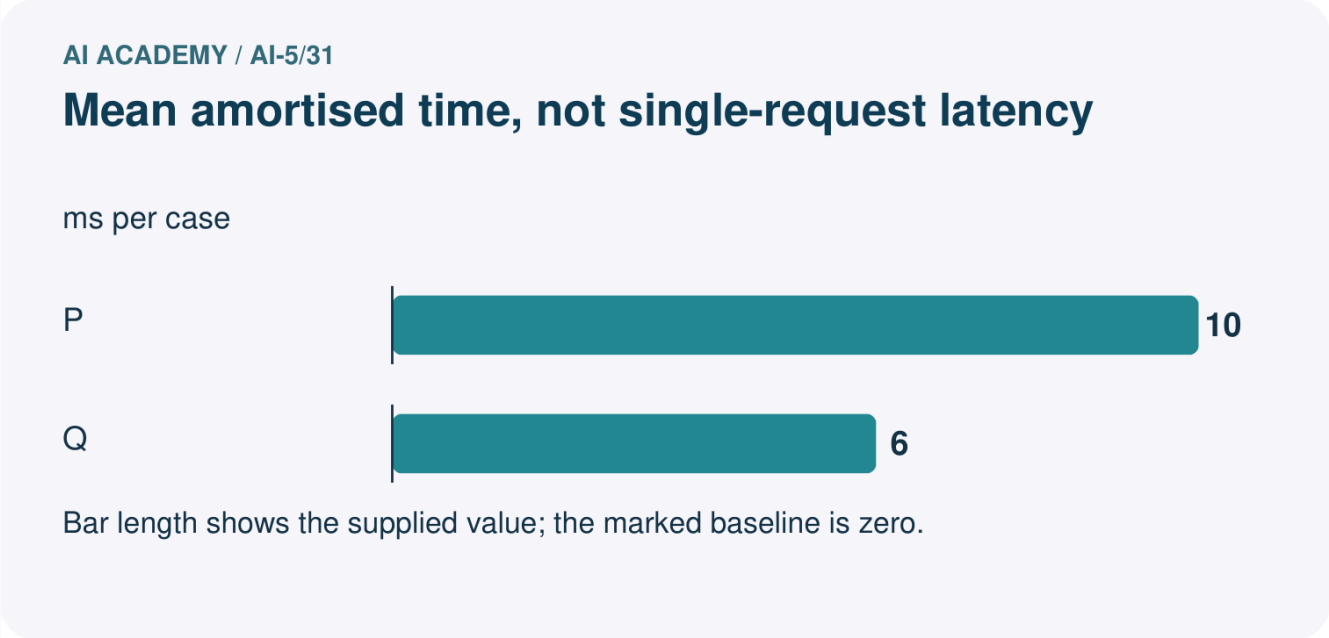
For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Efficiency and resource measurement

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/31

Efficiency and resource measurement

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Summarise P

Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both.

2. Summarise Q

Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality.

3. Choose eligible candidate

P meets the declared contract; Q is faster but ineligible.

CHANGE ONE THING

Lower the accuracy requirement to90% before a new selection exercise.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B’s time reduction relative to A and explain why the percentage **throughput** increase differs.

Teacher check: Teacher explanation: Only A qualifies; B misses the quality requirement. B reduces time by 8 / 20 = 40 percent. **Throughput** rises from 50 to about 83.33, an increase of about 66.67 percent relative to 50.

Time and **throughput** use different baseline denominators and are reciprocal for this fixed workload.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Efficiency and resource measurement. Proposed explanation: “Fewer parameters always means faster”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Hardware, operations, memory access, precision and implementation affect runtime; measure the declared workload.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

It divides the mean duration of a 40-case run by 40. Batch overhead and **process**ing overlap mean individual request **latency** needs its own measurement.

Changed-case check: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q’s measured accuracy did not improve.

Changed-case reasoning: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q’s measured accuracy did not improve.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use a units ledger: seconds per pass, examples per pass, milliseconds per example and examples per second. Draw every dense connection and one bias per **output**.

Extension: compare median-pass **throughput** with total examples divided by total elapsed time; explain why these can differ when pass durations vary. Neither alone measures a **latency** percentile.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve why shared examples and group proportions matter for comparisons.
7 to 18 minutes	Read the timing protocol and table. Label training, warm-up and inference work separately.
18 to 32 minutes	Complete W1 and trace the timing code, including milliseconds and examples per second.
32 to 45 minutes	Use W2 and the visual to apply the quality and time requirements before choosing a model.
Session 2 start to 8 minutes	Complete the guided alternative requirement without retroactively changing the original rule.
8 to 20 minutes	Count weights and biases and complete W3; discuss what memory and energy require beyond counts.
20 to 35 minutes	Collect W4 independently, requiring units, accuracy fractions, eligibility and parameter bytes.
35 to 45 minutes	Use W5 and the exit response to repair benchmark claims. Give feedback before the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main A/B medians 2/1.2 seconds, 20/12 ms, 50/83.33 throughput; original gate admits A only. Raw 4→3→2 parameters 23, bytes 92. Independent P/Q medians .5/.4 seconds, 10/8 ms, throughput 100/125; accuracy 47/50=.94 and 45/50=.90. At .92 quality and 12 ms only P qualifies. Independent 3→2→1 has weights 6+2=8, biases 2+1=3, total 11, raw bytes 44.

Require amortised-versus-request-latency distinction and no unsupported energy claim.

Diagnose the current misconceptions

Fewer parameters always means faster: Hardware, operations, memory access, precision and implementation affect runtime; measure the declared workload.

Total time divided by count is every request's **latency**: It is an amortised quantity. Individual requests can vary, and batching or concurrency changes waiting time.

Only the winning run consumed resources: Unsuccessful trials and warm-up also consume resources; record them in the appropriate experiment budget.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

Teacher explanation: P: median 0.5 s, 10 ms, 100 examples/s, accuracy 47 / 50 = 94 percent. Q: median 0.4 s, 8 ms, 125 examples/s, accuracy 45 / 50 = 90 percent. Only P qualifies.

The separate network has (3*2 + 2) + (2*1 + 1) = 11 parameters, or 44 raw bytes.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Summarise repeated timings

For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

Teacher explanation: A: 2.0 seconds, 20 ms per example, 50 examples per second. B: 1.2 seconds, 12 ms per example, about 83.33 examples per second.

The boundary covers 100 completed sequential **prediction**s on pre**process**ed **input**s, excluding loading and pre**process**ing. It does not establish worst-case or percentile request **latency**.

W2 Compare under requirements

With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

Teacher explanation: Only A qualifies; B misses the quality requirement. B reduces time by 8 / 20 = 40 percent. **Throughput** rises from 50 to about 83.33, an increase of about 66.67 percent relative to 50.

Time and **throughput** use different baseline denominators and are reciprocal for this fixed workload.

W3 Separate storage and memory

Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

Teacher explanation: There are 12 + 6 = 18 weights and 3 + 2 = 5 biases, totalling 23 parameters and 92 bytes. Activations, buffers, gradients, optimiser state and overhead can require additional memory.

Energy needs power-and-time **evidence**; parameter bytes alone provide neither peak **training** memory nor energy.

W5 Repair an unfair benchmark

A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

Teacher explanation: Control device, precision, workload, batch size, concurrency and timed boundary for a model-only comparison, or explicitly compare whole systems without attributing the result to the model alone.

Use the same quality cases, report repetitions and raw timings rather than one best trial, and count all **training** trials and warm-up work in their appropriate budgets.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: under the alternative rule set in advance, A and B both qualify. B wins the declared lower-amortised-time tie-breaker, 12 versus 20 ms.

Exit: best depends on requirements and B fails the original quality rule; greener is unsupported without energy **evidence**.

Use identical workload and declared conditions or label a whole-system comparison, measure completed work, repeat timings, retain raw results and quality outcomes, and reserve final evaluation after selection.

W1 Summarise repeated timings

W2 Compare under requirements

W3 Separate storage and memory

W4 Evaluate a new case independently

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

Teacher explanation: P: median 0.5 s, 10 ms, 100 examples/s, accuracy 47 / 50 = 94 percent. Q: median 0.4 s, 8 ms, 125 examples/s, accuracy 45 / 50 = 90 percent. Only P qualifies.

The separate network has (3*2 + 2) + (2*1 + 1) = 11 parameters, or 44 raw bytes.

W5 Repair an unfair benchmark

Independent assessment and fresh retry

W4 requires P: 0.5 s, 10 ms, 100/s, 94 percent; Q: 0.4 s, 8 ms, 125/s, 90 percent; only P eligible; 11 parameters and 44 bytes.

Fresh retry: 40 shared cases, R times 0.8, 1.0, 0.6 seconds with 38 correct; S times 0.4, 0.6, 0.5 with 36 correct. Require 95 percent accuracy and at most 25 ms.

R gives 0.8 s, 20 ms, 50/s and 95 percent; S gives 0.5 s, 12.5 ms, 80/s and 90 percent. Only R qualifies. A 2 to 3 to 1 dense network has 13 parameters and 52 bytes at four bytes each.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled efficiency and resource measurement **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **throughput** and amortised time per example from repeated timings.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a units ledger: seconds per pass, examples per pass, milliseconds per example and examples per second. Draw every dense connection and one bias per **output**.

Extension: compare median-pass **throughput** with total examples divided by total elapsed time; explain why these can differ when pass durations vary. Neither alone measures a **latency** percentile.

Week 32 documents a network experiment, preserving methods, data splits, versions, raw results, resource measurements and limitations.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/32, ai-5/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Calculate **throughput** and amortised time per example from repeated timings

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies?

Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

Reasoned practice answer: Teacher explanation: Only A qualifies; B misses the quality requirement. B reduces time by 8 / 20 = 40 percent.

Throughput rises from 50 to about 83.33, an increase of about 66.67 percent relative to 50. Time and **throughput** use different baseline denominators and are reciprocal for this fixed workload.

Check every learner before discussion: Use the worked case for Efficiency and resource measurement. Proposed explanation: "Fewer parameters always means faster". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Hardware, operations, memory access, precision and implementation affect runtime; measure the declared workload.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Lower the accuracy requirement to90% before a new selection exercise. Require a **prediction** and a reason before comparing. Expected result: Both become eligible; Q has the smaller measured mean amortised time.

Reason: The acceptance contract changed; Q's measured accuracy did not improve.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

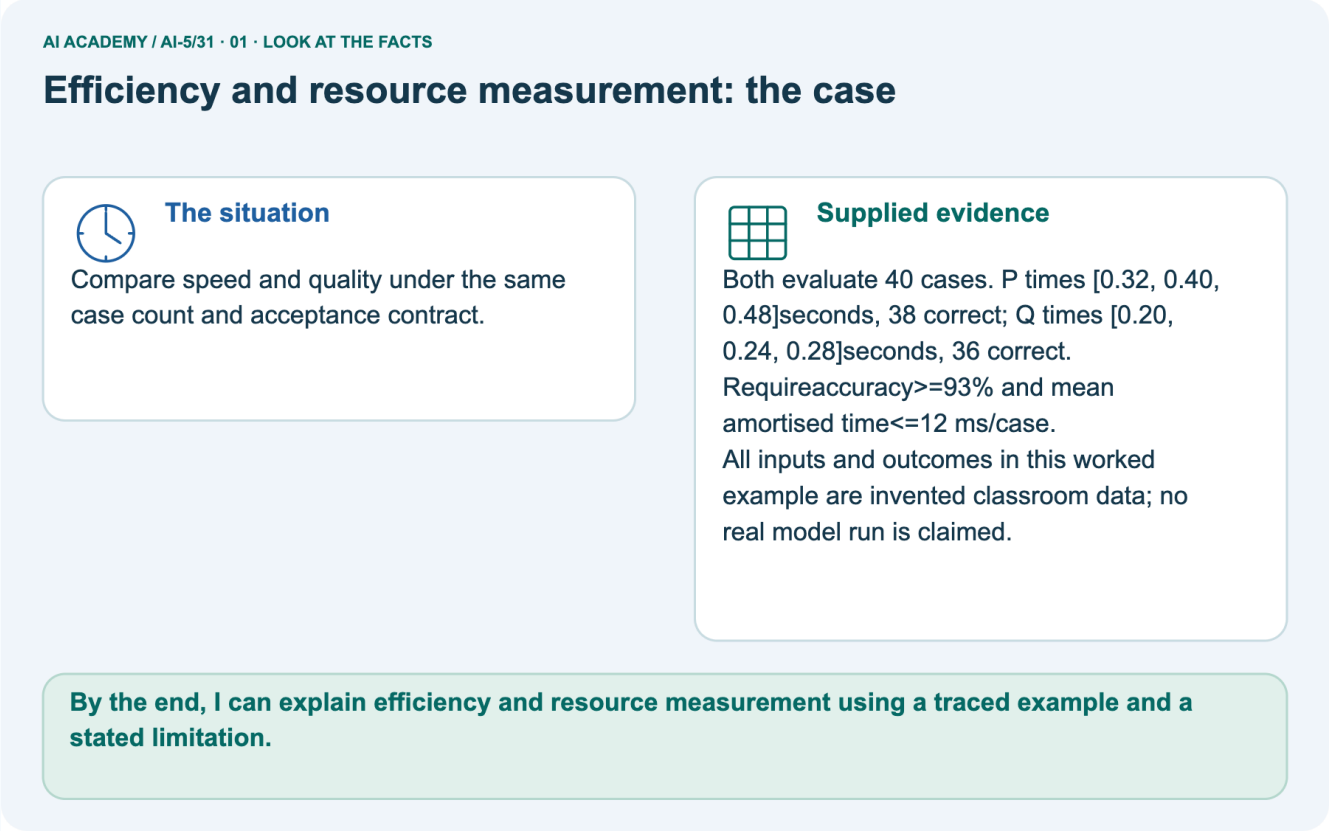
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Efficiency and resource measurement: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

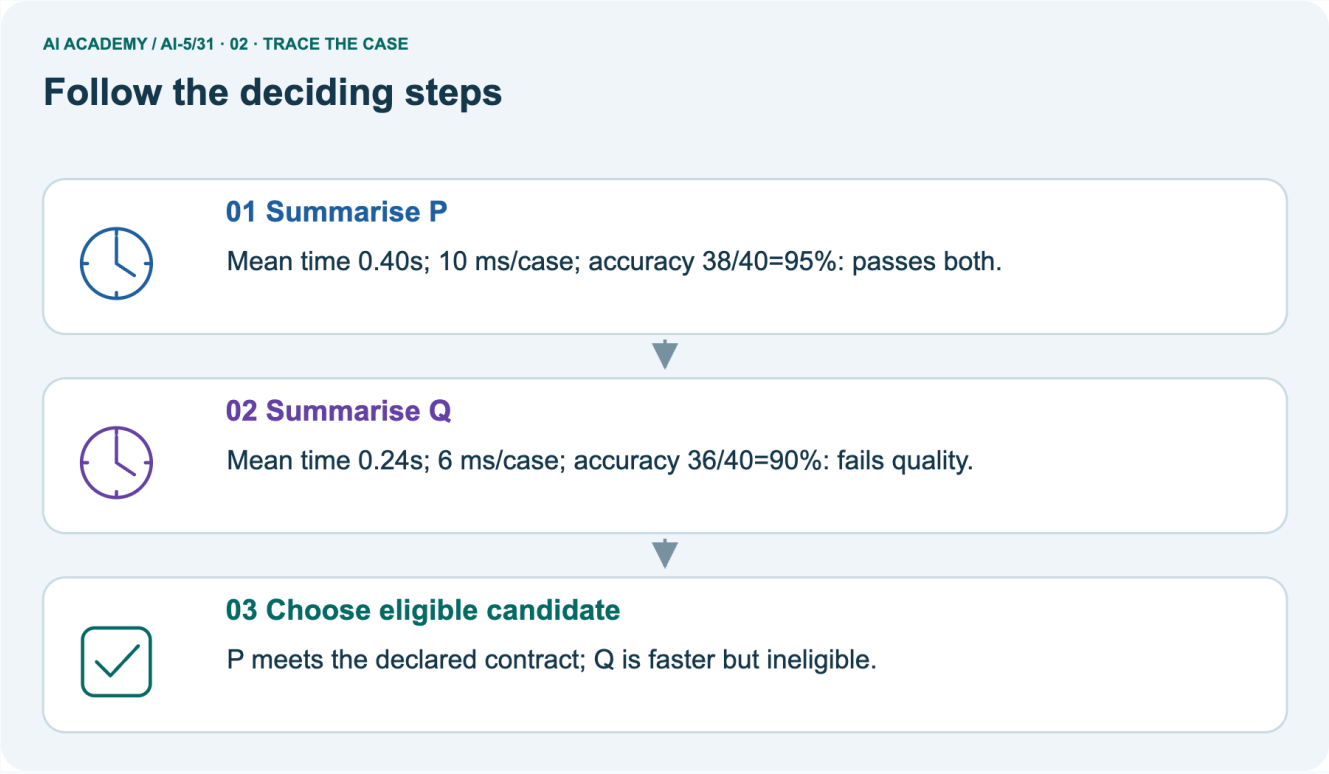
Compare speed and quality under the same case count and acceptance contract. Supplied facts: Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct.

Requireaccuracy>=93% and mean amortised time<=12 ms/case. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



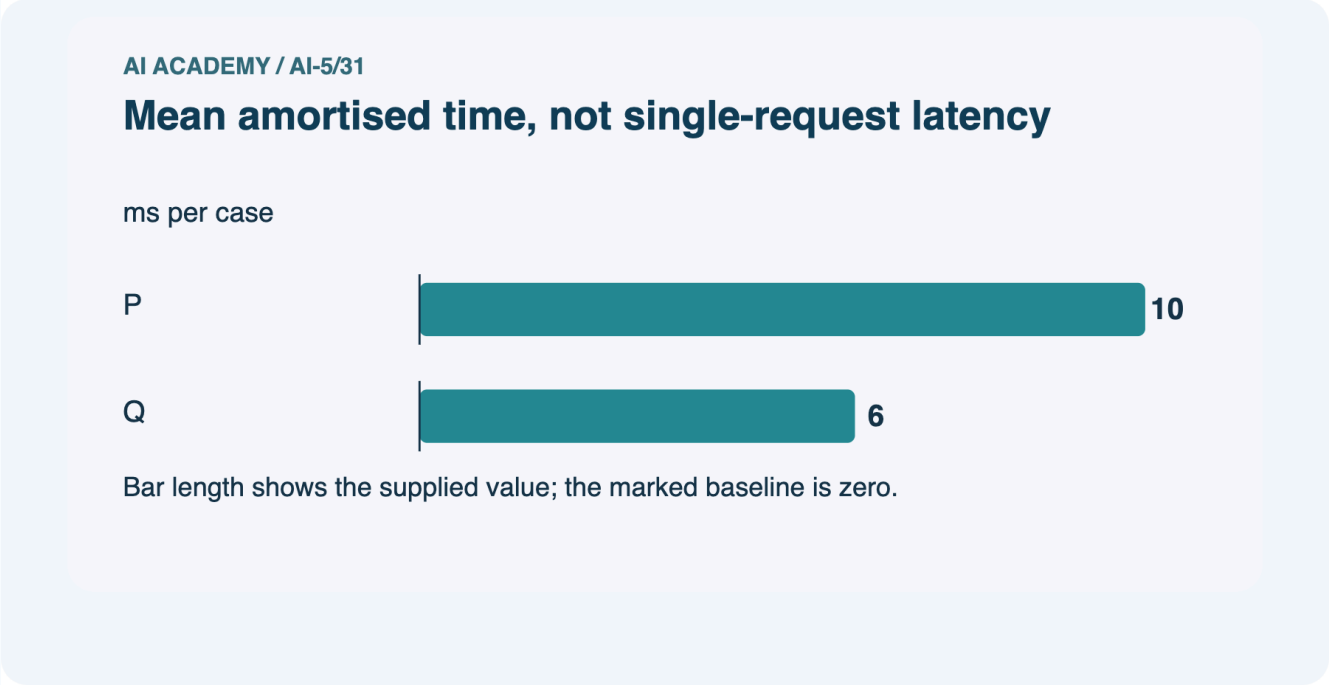
Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise P: Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both. → Summarise Q: Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality. → Choose eligible candidate: P meets the declared contract; Q is faster but ineligible.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Mean amortised time, not single-request latency



Follow the labelled picture with the matching explanation. Open the vector diagram

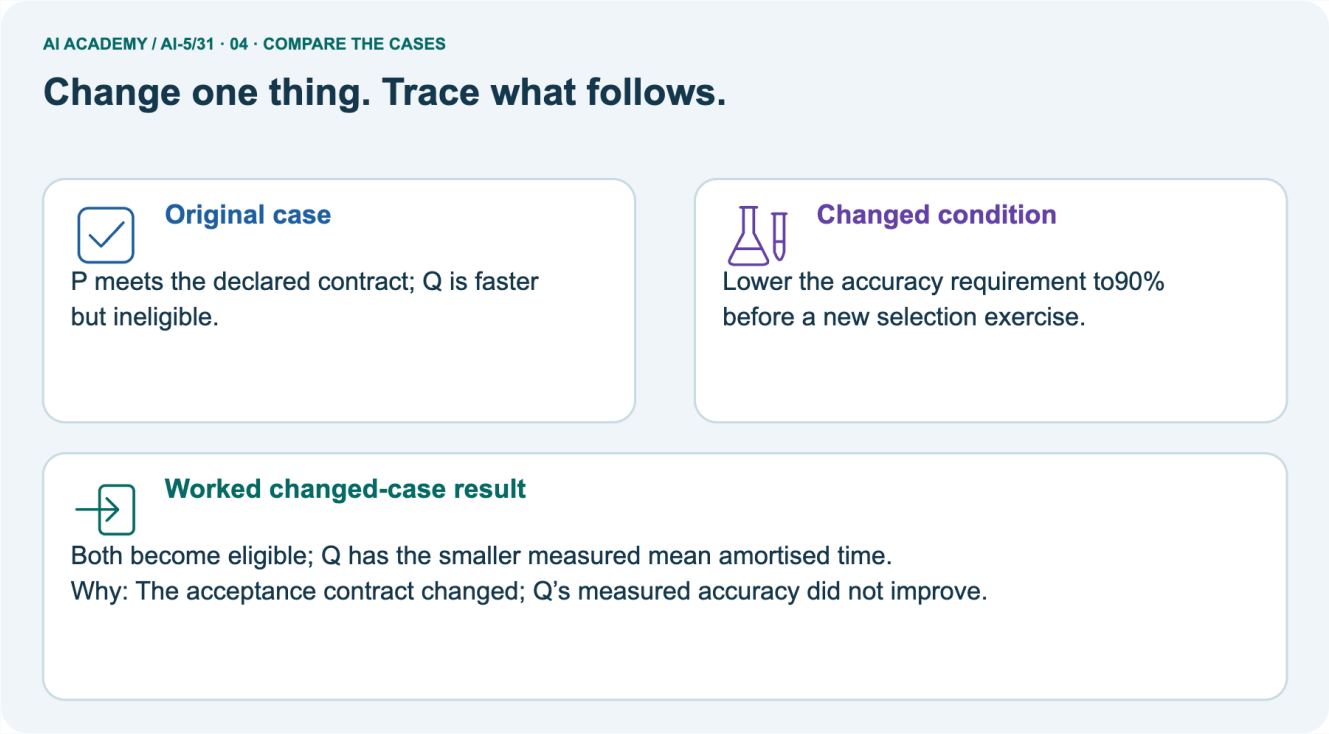
Mean amortised time, not single-request **latency**. Units: ms per case. P: 10. Q: 6.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: P meets the declared contract; Q is faster but ineligible. Changed condition: Lower the accuracy requirement to90% before a new selection exercise.

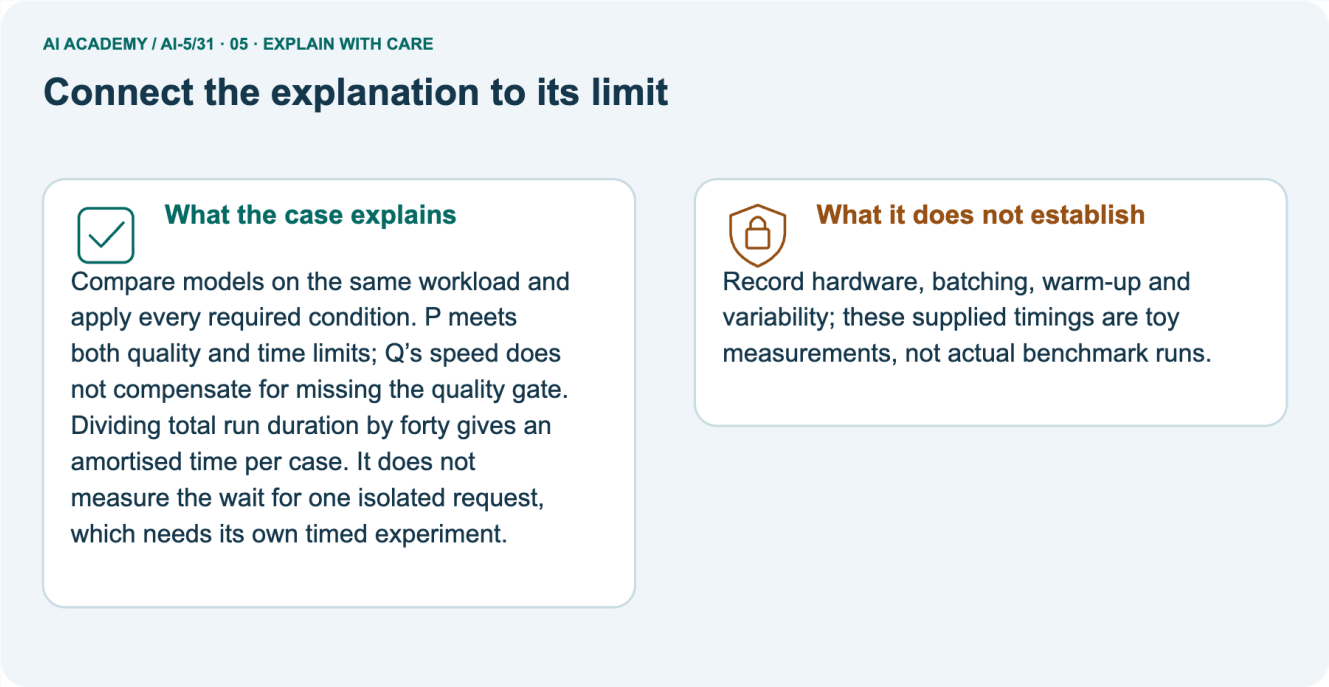
Expected result: Both become eligible; Q has the smaller measured mean amortised time. Reason: The acceptance contract changed; Q's measured accuracy did not improve.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The acceptance contract changed; Q's measured accuracy did not improve.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

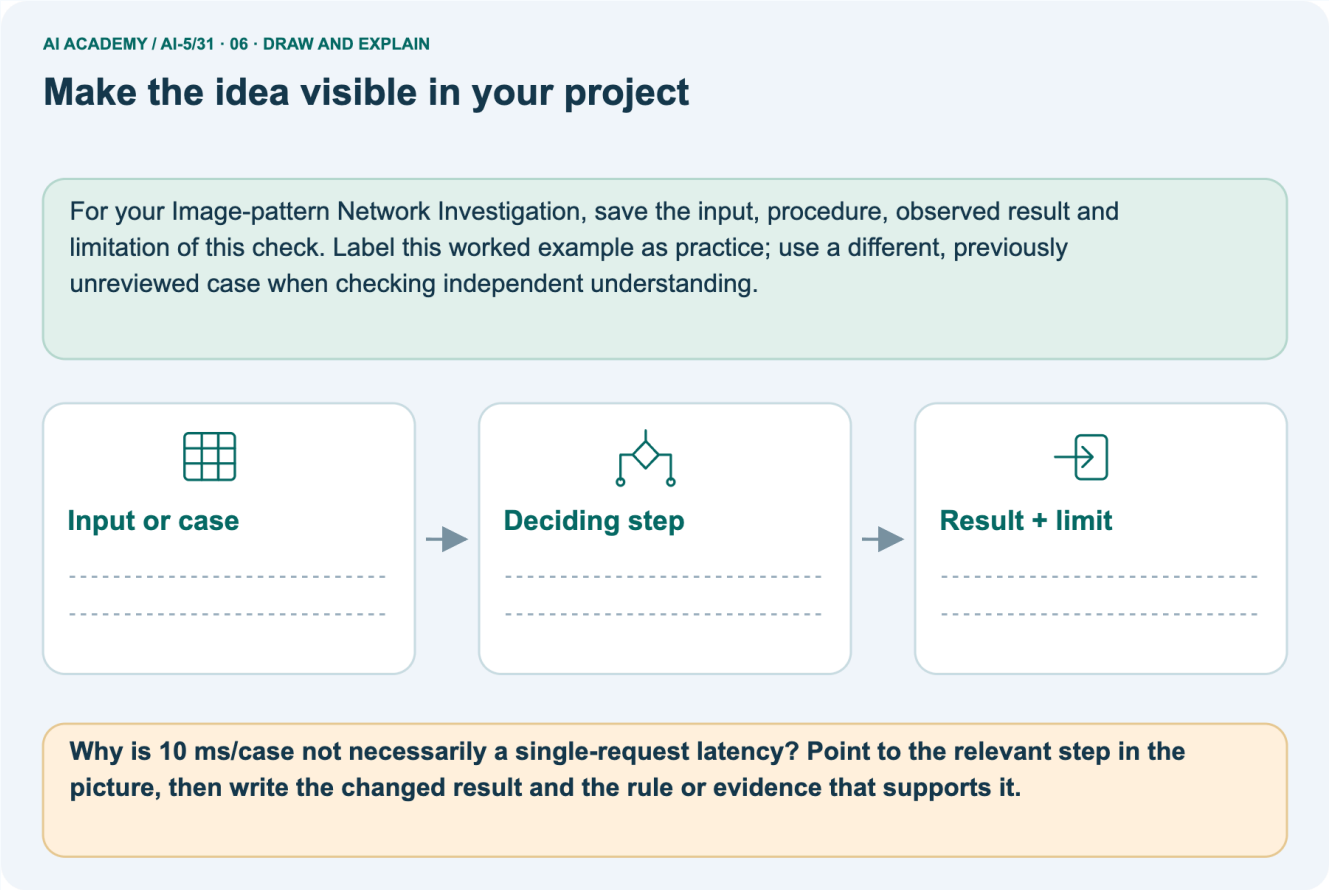
Limit: Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It divides the mean duration of a 40-case run by 40. Batch overhead and **process**ing overlap mean individual request **latency** needs its own measurement.

Changed-case check: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q's measured accuracy did not improve.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Efficiency and resource measurement: the case

02. Follow the deciding steps

03. Mean amortised time, not single-request latency

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.